

The left side of the slide features a series of vertical stripes in various shades of gray and blue. Overlaid on these stripes are several dark blue circles of different sizes, resembling bubbles. One of these bubbles contains the number '1'.

DATA MODELING USING THE ENTITY-RELATIONSHIP MODEL

1

MAJOR COMPONENTS

- Entities
- Attributes
- Relationships



ENTITY

- Term used to mean three different meanings
 - Entity type
 - Entity instance
 - Entity set



ENTITY TYPE

- A name/label assigned to items/objects that exist in an environment and that have similar properties
- It could be person, place, event or even concept



ENTITY INSTANCE & SET

- A particular object belonging to a particular entity type
- Entity Type: Employee
- Entity Instance: M. Sharif
- Entity Set: All employees



TYPES OF ENTITY TYPES

- Entity types can be classified into regular/strong/independent ETs or weak/dependent ETs



WEAK ENTITY TYPES

- An entity type whose instances cannot exist without being linked with instances of some other entity type, i.e., they cannot exist independently



STRONG ENTITY TYPE

- A strong/regular entity type is the one whose instances can exist independently, i.e., without being linked to other instances
- Strong ETs have their own identity

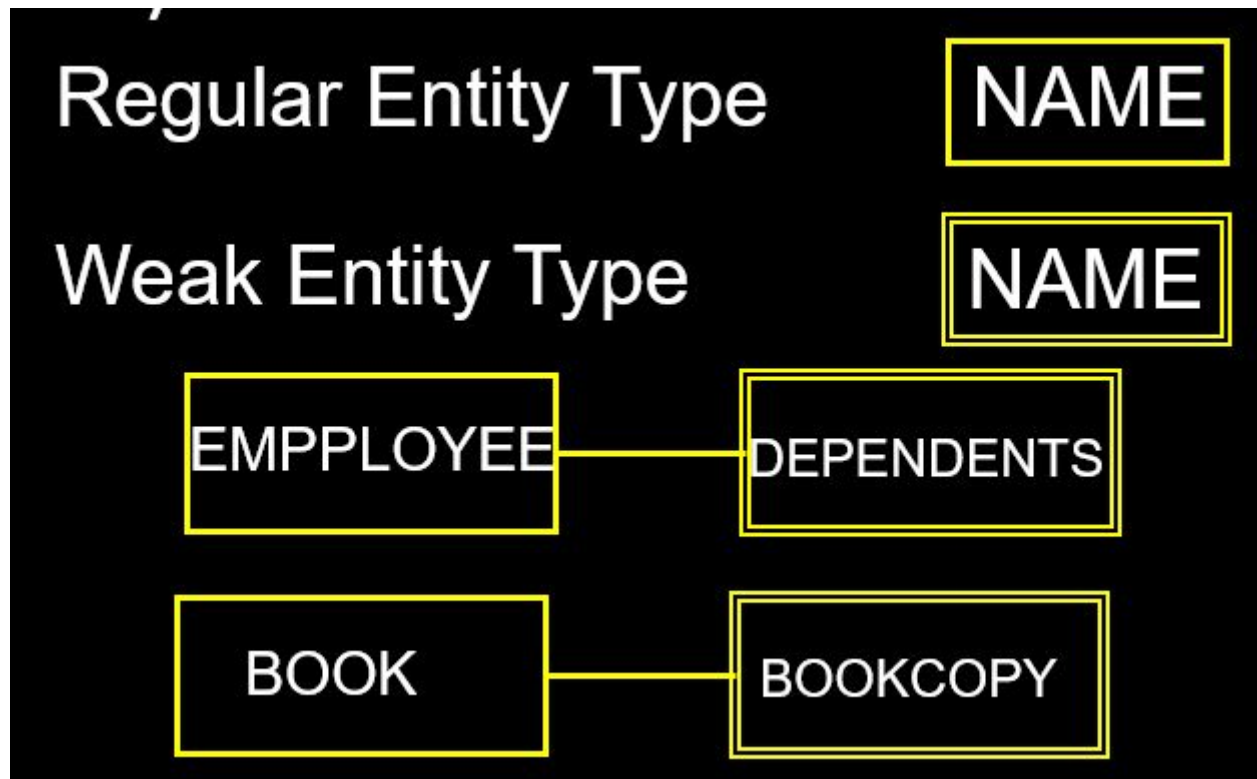


NAMING ENTITY TYPES

- Singular noun recommended
- Organization specific names
- Write in capitals
- Abbreviations can be used, be consistent



SYMBOLS



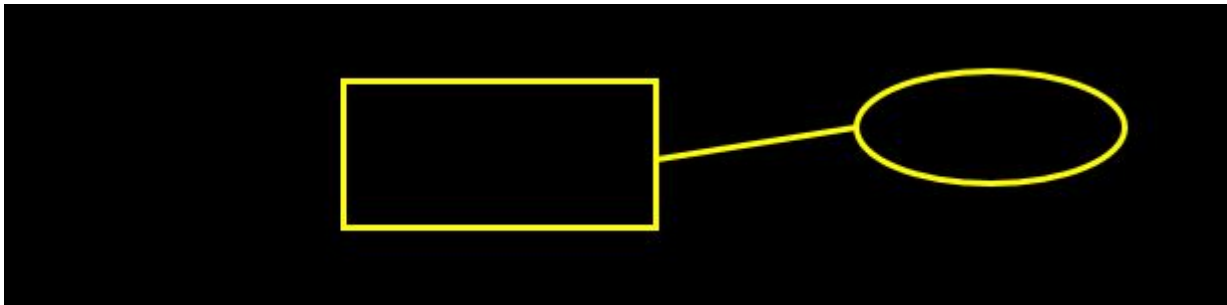
ER MODEL CONCEPTS

□ Attributes

- They are properties used to describe an entity.
- Each attribute has a data type
- E.g. integer, string, subrange, enumerated type, ...

SYMBOLS FOR ATTRIBUTES

- Each represented as an oval, linked with an ET symbol



TYPES OF ATTRIBUTES

- Single vs composite
- Single valued vs multi-valued
- Stored vs derived
- Complex attribute



TYPES OF ATTRIBUTES

□ Simple

- Each entity has a single atomic value for the attribute. For example, SSN or Sex.

□ Composite

- The attribute is composed of several components. For example:
 - Address(House#, Street, City, State, ZipCode, Country),
 - Name(FirstName, MiddleName, LastName).

□ Multi-valued

- An entity may have multiple values for that attribute. For example:
- PreviousDegrees of a STUDENT.

□ Stored and Derived

TYPES OF ATTRIBUTES

- Complex attribute - The composite and multi-valued attributes may be nested arbitrarily to any number of levels,
 - For example, PreviousDegrees of a STUDENT is a composite multi-valued attribute denoted by {PreviousDegrees (College, Year, Degree, Field)}
 - Multiple PreviousDegrees values can exist
 - Each has four subcomponent attributes:
 - College, Year, Degree, Field

EXAMPLE OF A COMPOSITE ATTRIBUTE

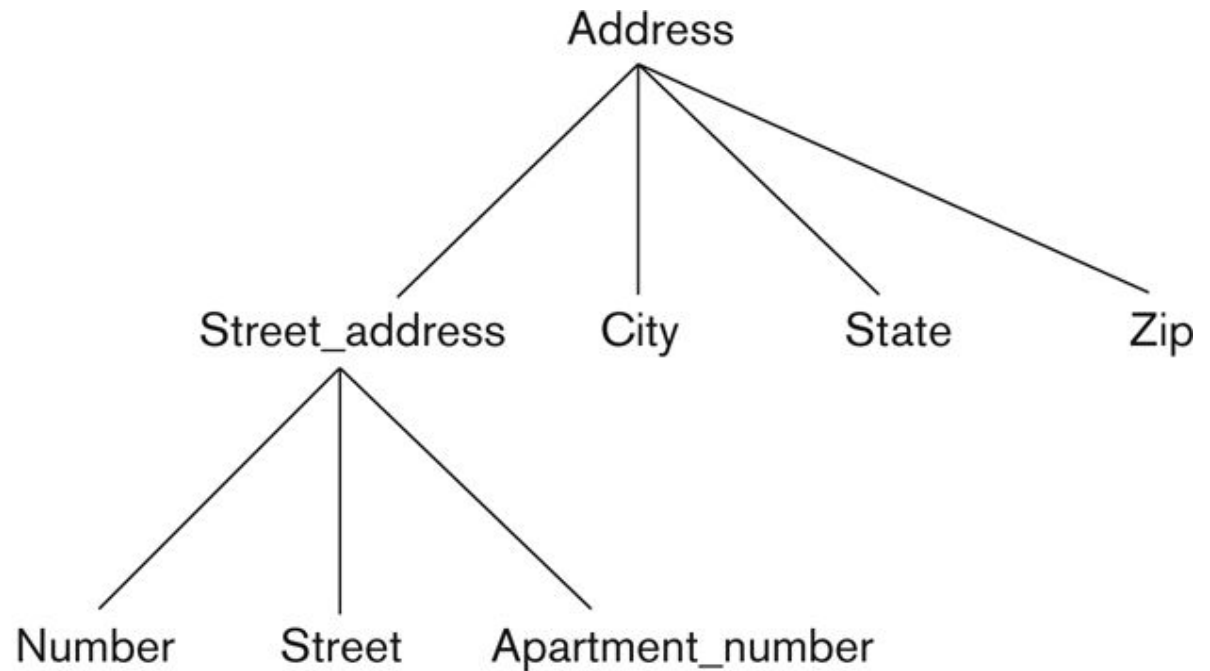
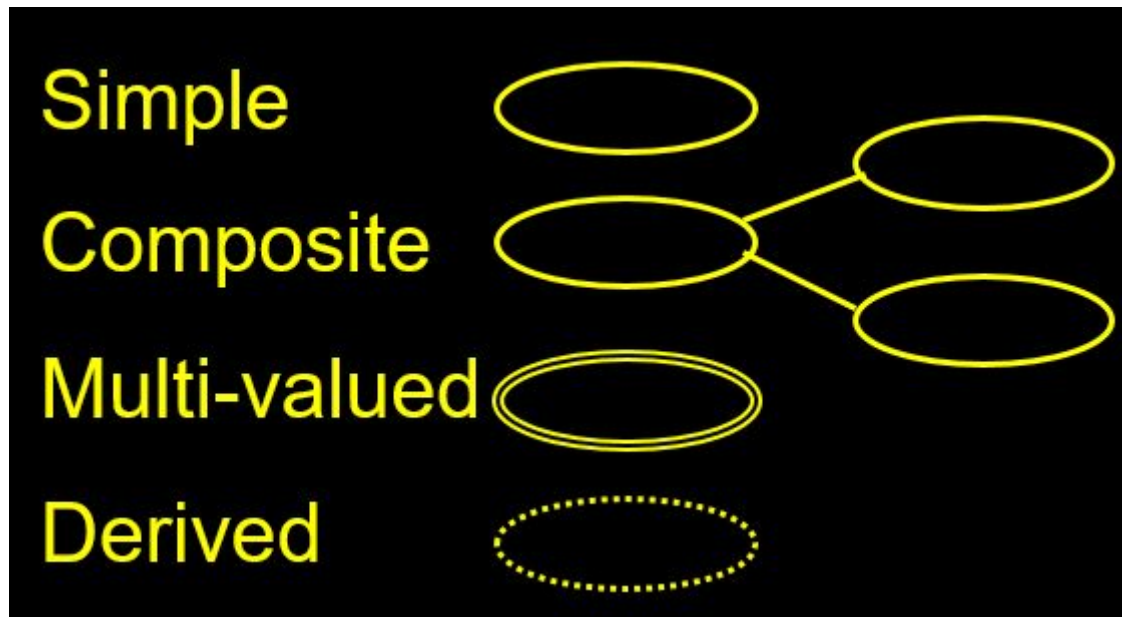


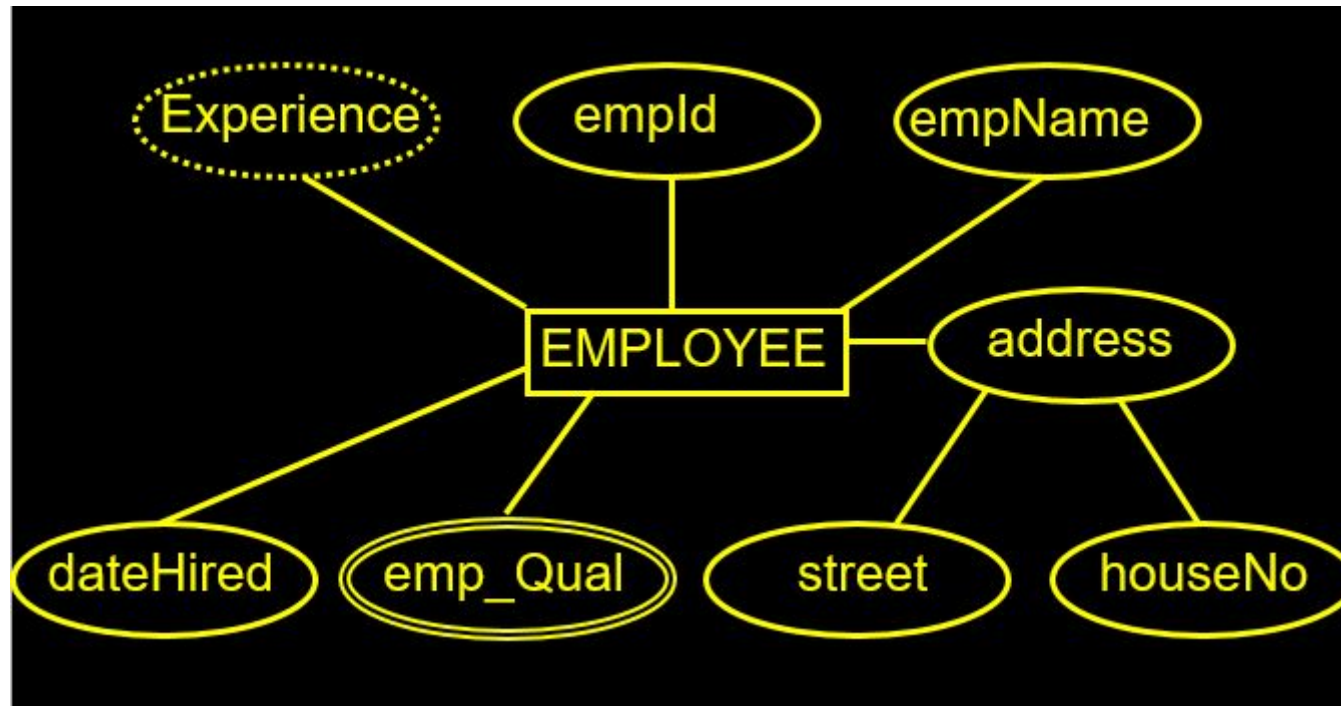
Figure 3.4

A hierarchy of composite attributes.

SYMBOLS FOR ATTRIBUTES



EXAMPLE



ENTITY TYPES AND KEY ATTRIBUTES

- Entities with the same basic attributes are grouped into an Entity type.
 - For example, the entity type EMPLOYEE and PROJECT.

- Key attribute
 - For example, SSN of EMPLOYEE.

KEY ATTRIBUTES

- A key attribute may be composite.
 - VehicleTagNumber is a key of the CAR entity type with components (Number, State).
- An entity type may have more than one key.
 - The CAR entity type may have two keys:
 - VehicleId (popularly called VIN)
 - VehicleTagNumber (Number, State), aka license plate number.
- Each key is underlined

ENTITY TYPE CAR WITH TWO KEYS AND A CORRESPONDING ENTITY SET

(a)

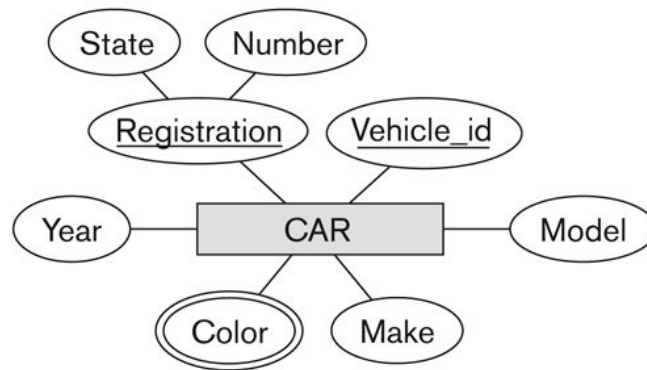


Figure 3.7

The CAR entity type with two key attributes, Registration and Vehicle_id. (a) ER diagram notation. (b) Entity set with three entities.

(b)

CAR

Registration (Number, State), Vehicle_id, Make, Model, Year, {Color}

CAR₁

((ABC 123, TEXAS), TK629, Ford Mustang, convertible, 2004 {red, black})

CAR₂

((ABC 123, NEW YORK), WP9872, Nissan Maxima, 4-door, 2005, {blue})

CAR₃

((VSY 720, TEXAS), TD729, Chrysler LeBaron, 4-door, 2002, {white, blue})

⋮

Entity set is the current *state* of the entities that is stored in the database

INITIAL DESIGN OF ENTITY TYPES: EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT

Entity types in the
COMPANY database:
DEPARTMENT
PROJECT
EMPLOYEE
DEPENDENT

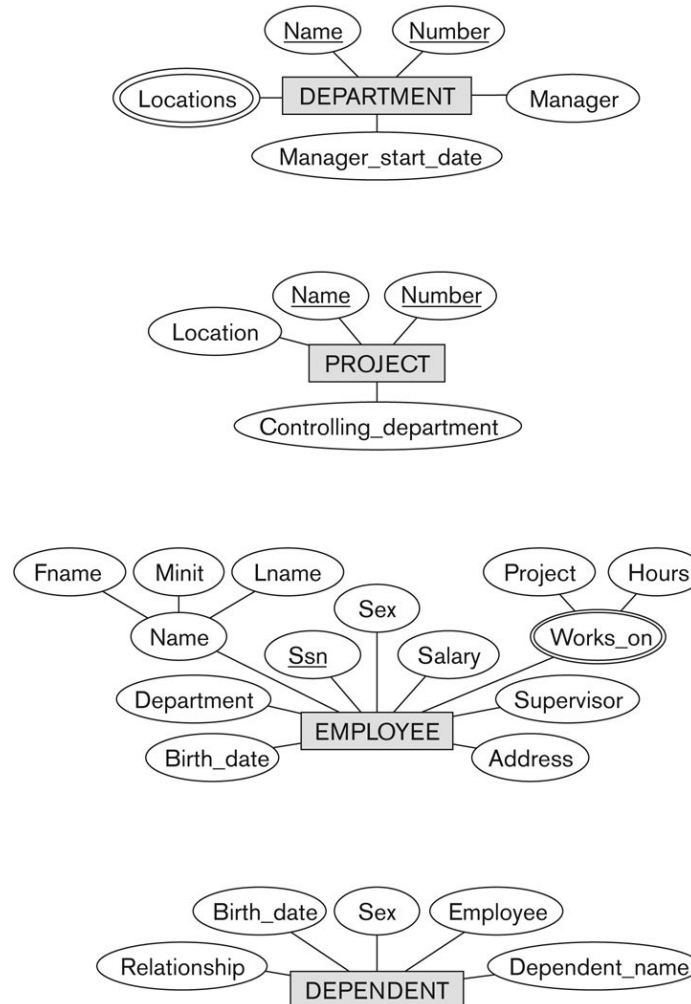


Figure 3.8
Preliminary design of entity
types for the COMPANY
database. Some of the
shown attributes will be
refined into relationships.

REFINING THE INITIAL DESIGN BY INTRODUCING RELATIONSHIPS

- Some aspects in the requirements will be represented as **relationships**

- ER model has three main concepts:
 - Entities (and their entity types and entity sets)
 - Attributes (simple, composite, multivalued)
 - Relationships (and their relationship types and relationship sets)

RELATIONSHIPS

- A **relationship** relates two or more distinct entities with a specific meaning.
 - For example, EMPLOYEE John *works on* the ProductX PROJECT,
 - EMPLOYEE Franklin *manages* the Research DEPARTMENT.
- Relationships of the same type are grouped into a **relationship type**.
- Degree of a relationship type is the no of participating entity types.
 - Both MANAGES and WORKS_ON are *binary* relationships.

RELATIONSHIP INSTANCES OF THE WORKS_FOR RELATIONSHIP BETWEEN EMPLOYEE AND DEPARTMENT

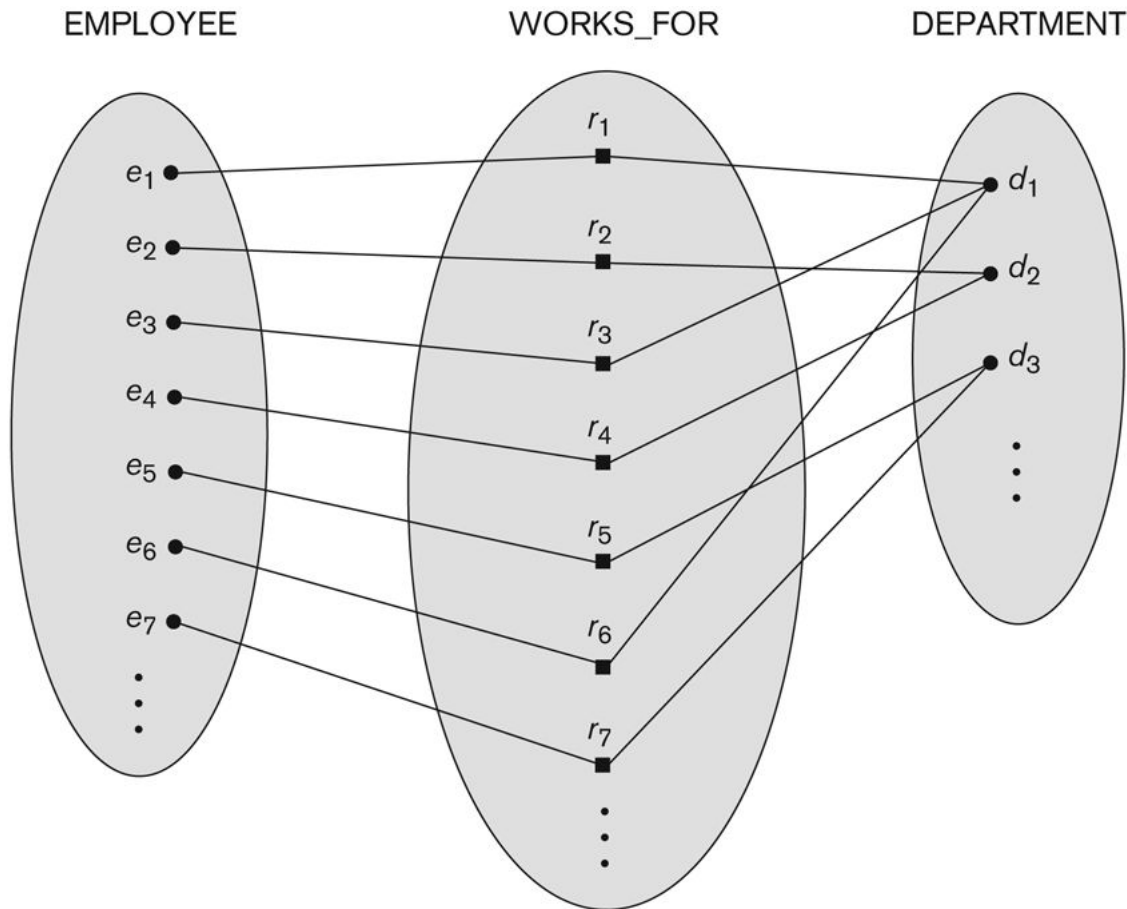


Figure 3.9

Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

RELATIONSHIP INSTANCES OF THE WORKS_ON RELATIONSHIP BETWEEN EMPLOYEE AND PROJECT

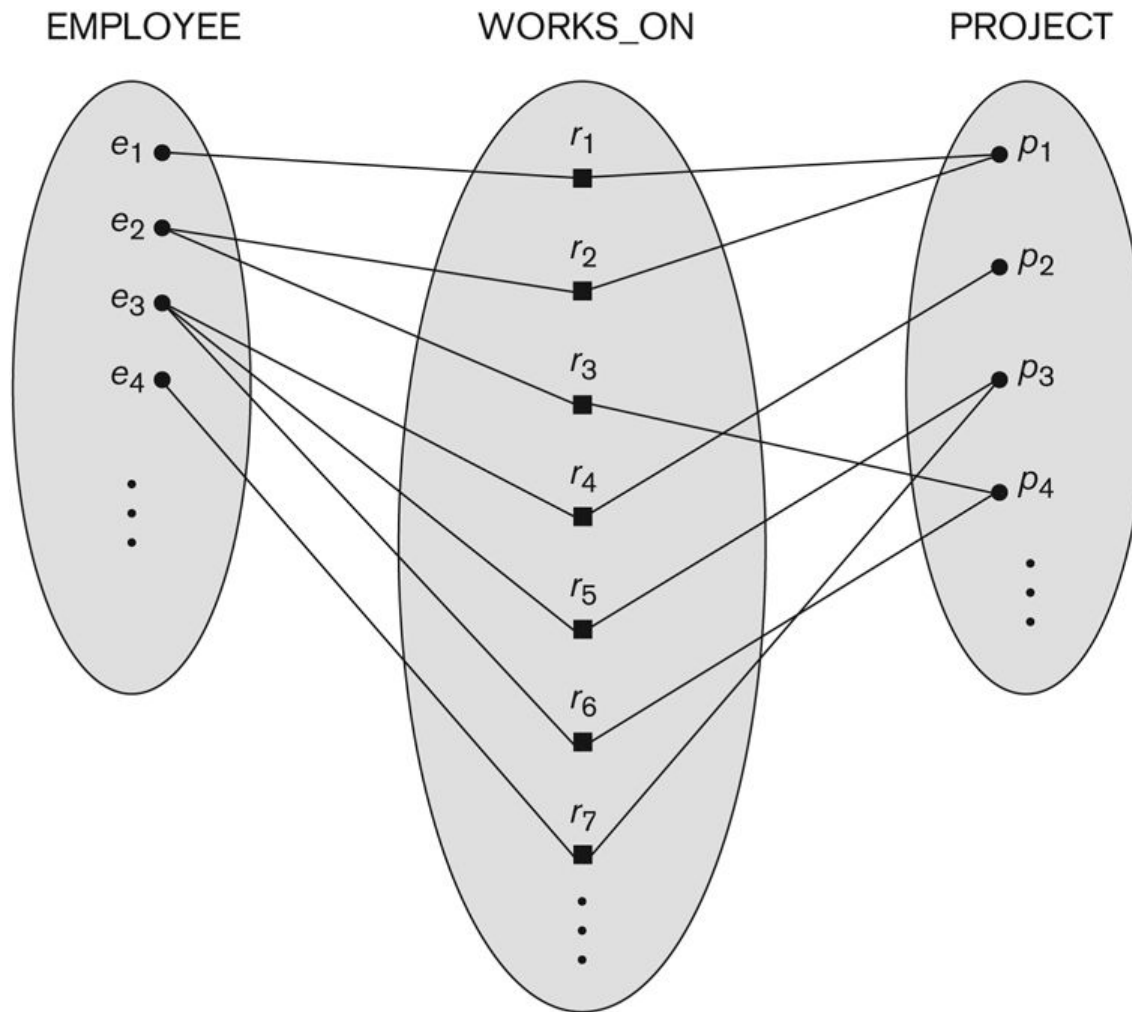
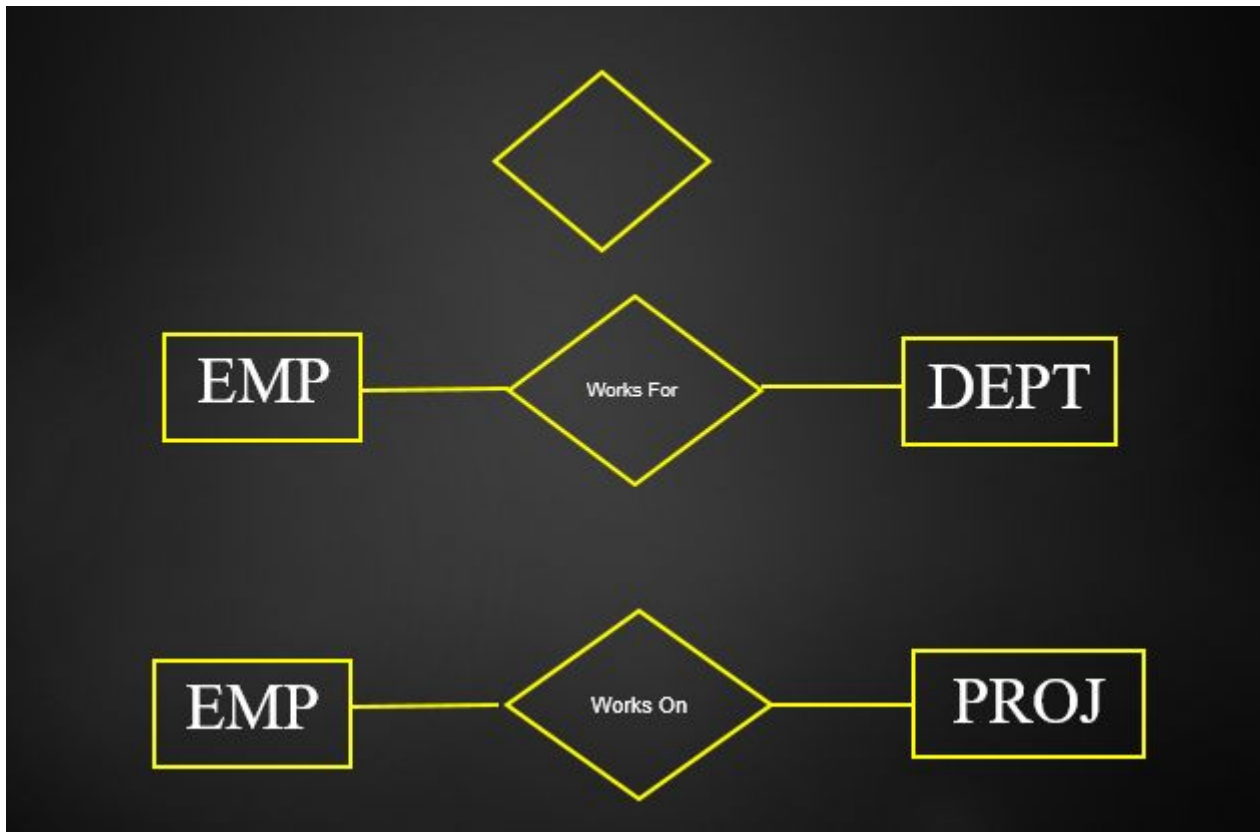


Figure 3.13
An M:N relationship,
WORKS_ON.

SYMBOL



RELATIONSHIP TYPE VS. RELATIONSHIP SET

□ Relationship Set:

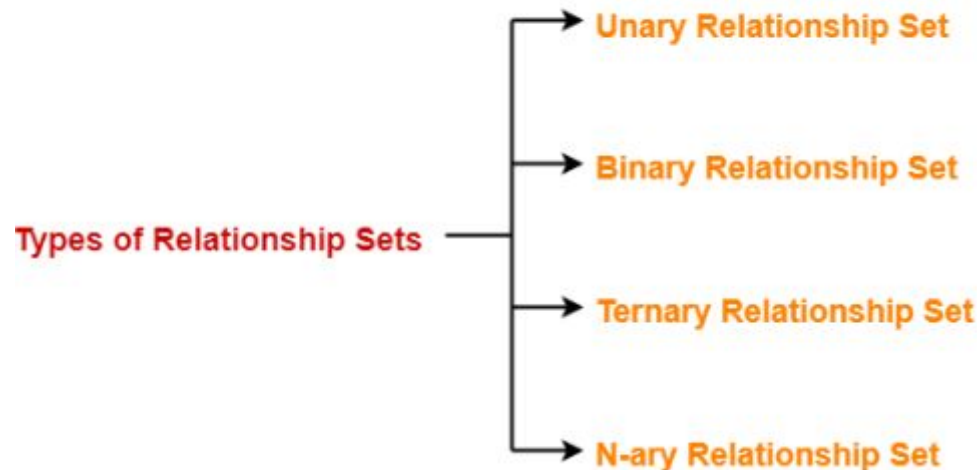
- A relationship set is a set of relationships of same type.

□ Relationship Type:

- Identifies the relationship name and the participating entity types
- Also identifies certain relationship constraints

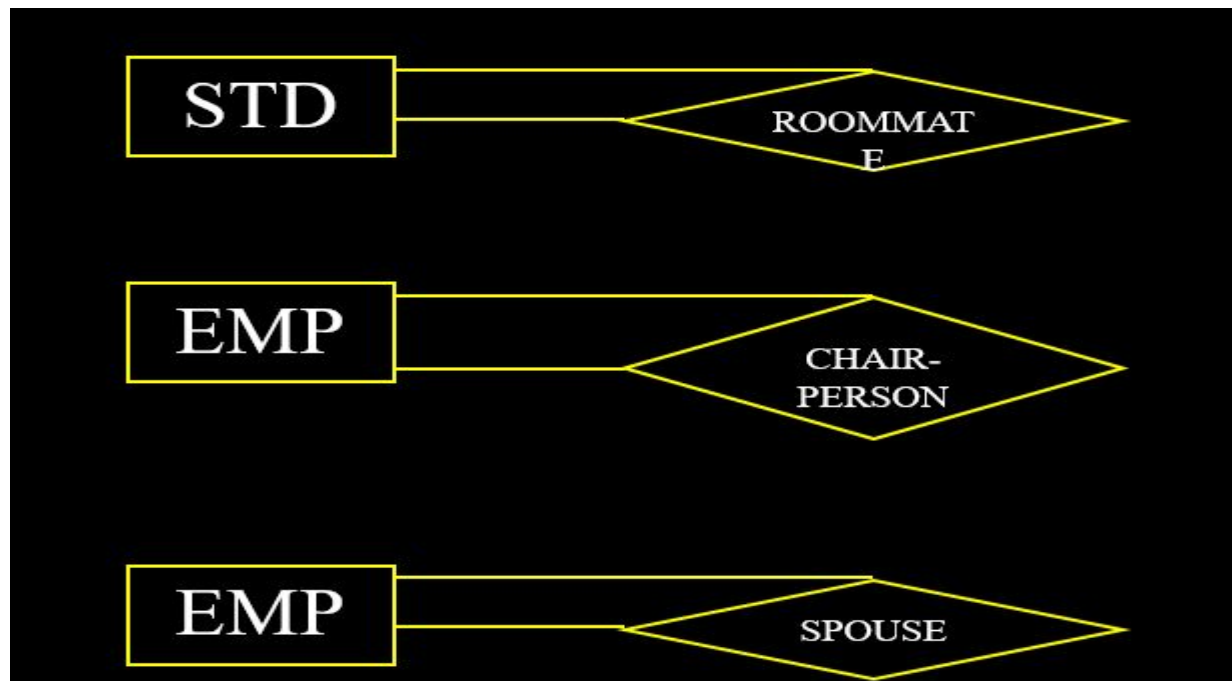
TYPES OF RELATIONSHIP SET

- The number of entity sets that participate in a relationship set is termed as the degree of that relationship set. Thus



UNARY RELATIONSHIP

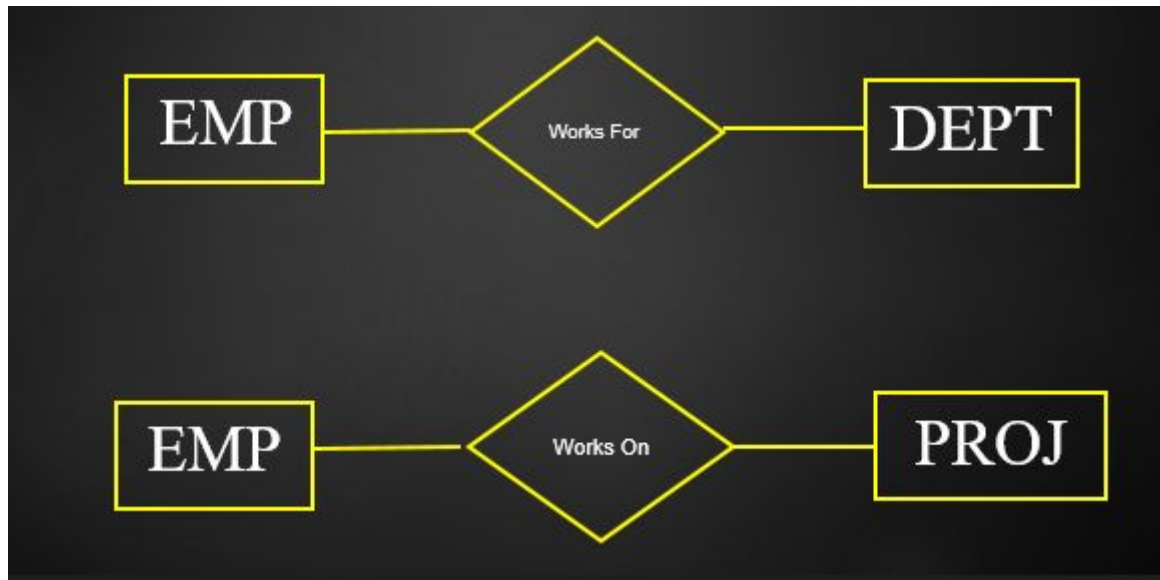
- An ET linked with itself, also called recursive relationship
- Example Roommate, where STUDENT is linked with STUDENT



COMPANY DATABASE: RELATIONSHIPS

- Six relationship types are identified
- All are *binary* relationships(degree 2)
 - WORKS_FOR (between EMPLOYEE, DEPARTMENT)
 - MANAGES (between EMPLOYEE, DEPARTMENT)
 - CONTROLS (between DEPARTMENT, PROJECT)
 - WORKS_ON (between EMPLOYEE, PROJECT)
 - SUPERVISION (between EMPLOYEE (as subordinate), EMPLOYEE (as supervisor))
 - DEPENDENTS_OF (between EMPLOYEE, DEPENDENT)

EXAMPLE



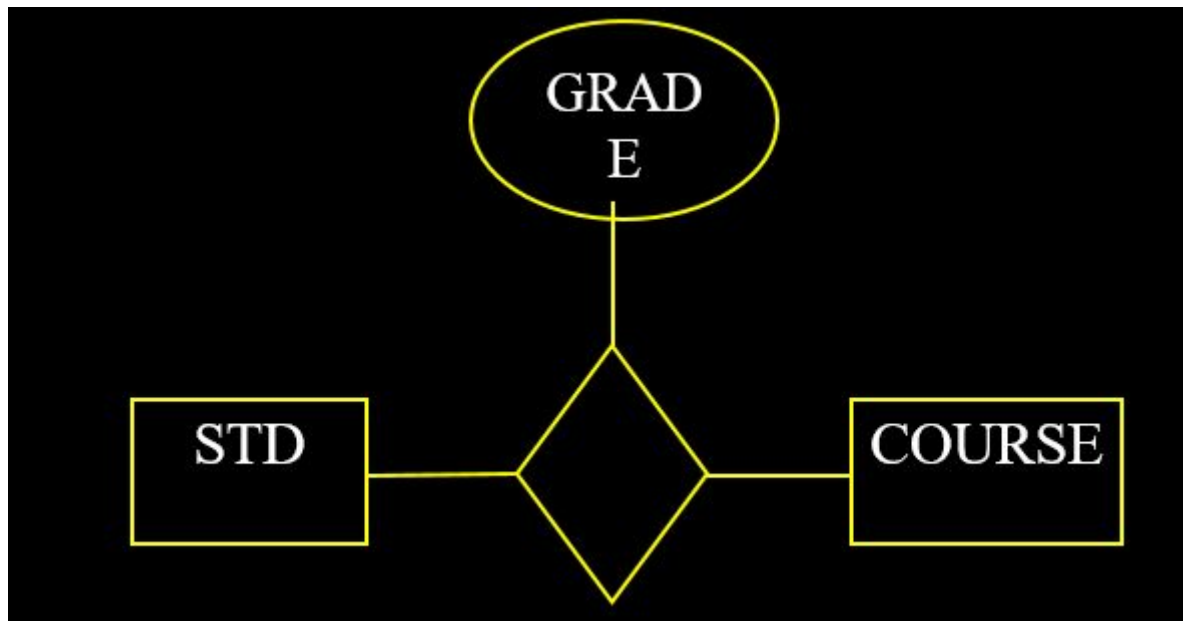
BINARY RELATIONSHIPS

- May also have instances, that can be formally described in an ordered pair form
- Employee and Project
- $\{(E1001, A), (E1020, B), (E1002, B), (E1058, C)\}$



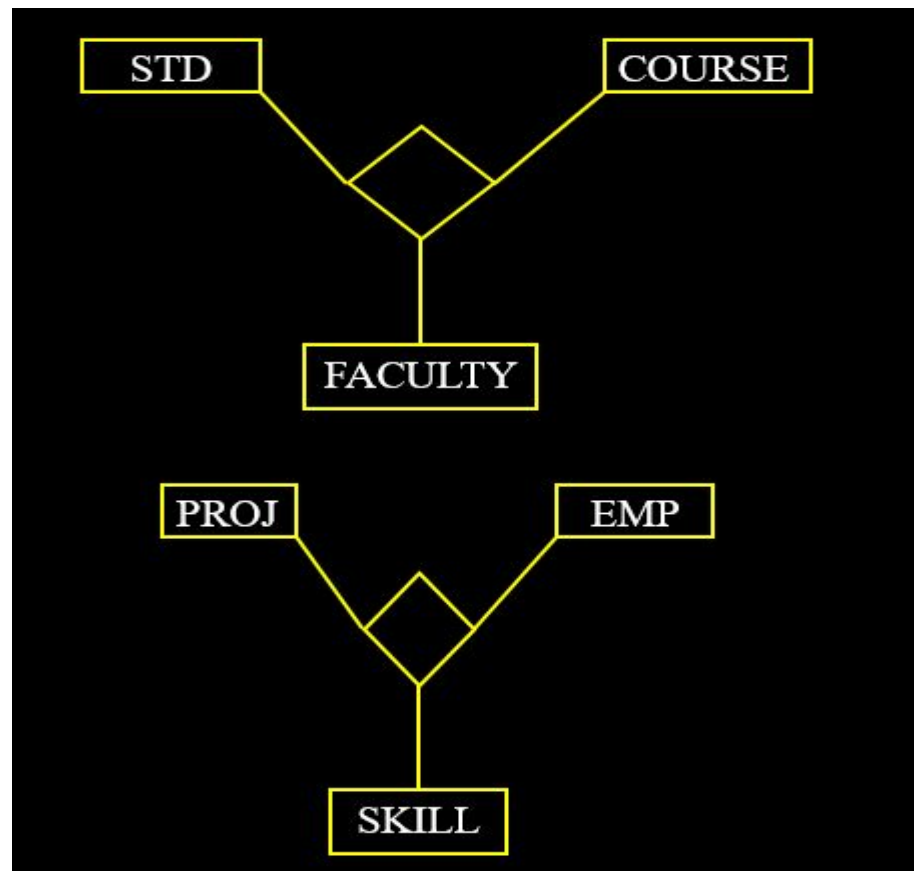
ATTRIBUTES OF THE RELATIONSHIPS

- The key (combination of key STD+Course)
- The relationships can have their descriptive attributes
- Where to place



TERNARY RELATIONSHIPS

- One that involves three entity types
- STUDENT-CLASS-FACULTY

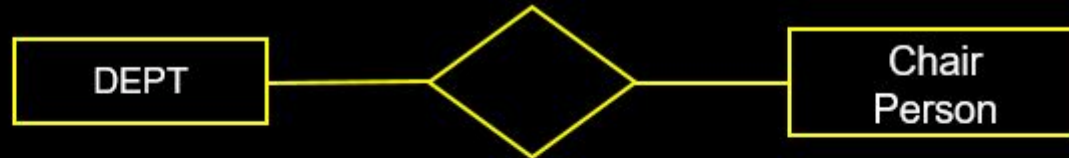


CONSTRAINTS ON RELATIONSHIPS TYPE

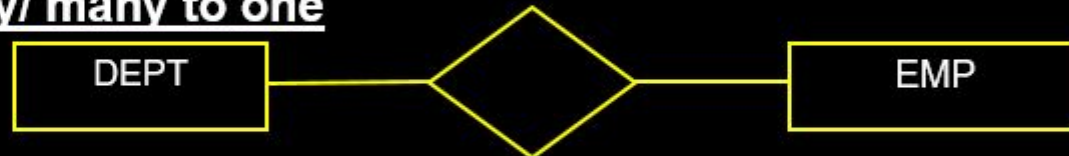
- Constraints on Relationship Types (ratio constraints)
 - Cardinality Ratio (specifies *maximum* participation)
 - One-to-one (1:1)
 - One-to-many (1:N) or Many-to-one (N:1)
 - Many-to-many (M:N)
 - Existence Dependency Constraint (specifies *minimum* participation) (also called participation constraint)
 - zero (optional participation, not existence-dependent)
 - one or more (mandatory participation, existence-dependent)

TYPES OF CARDINALITIES

One to one



One to many/ many to one



Many to many



MANY-TO-ONE (N:1) RELATIONSHIP

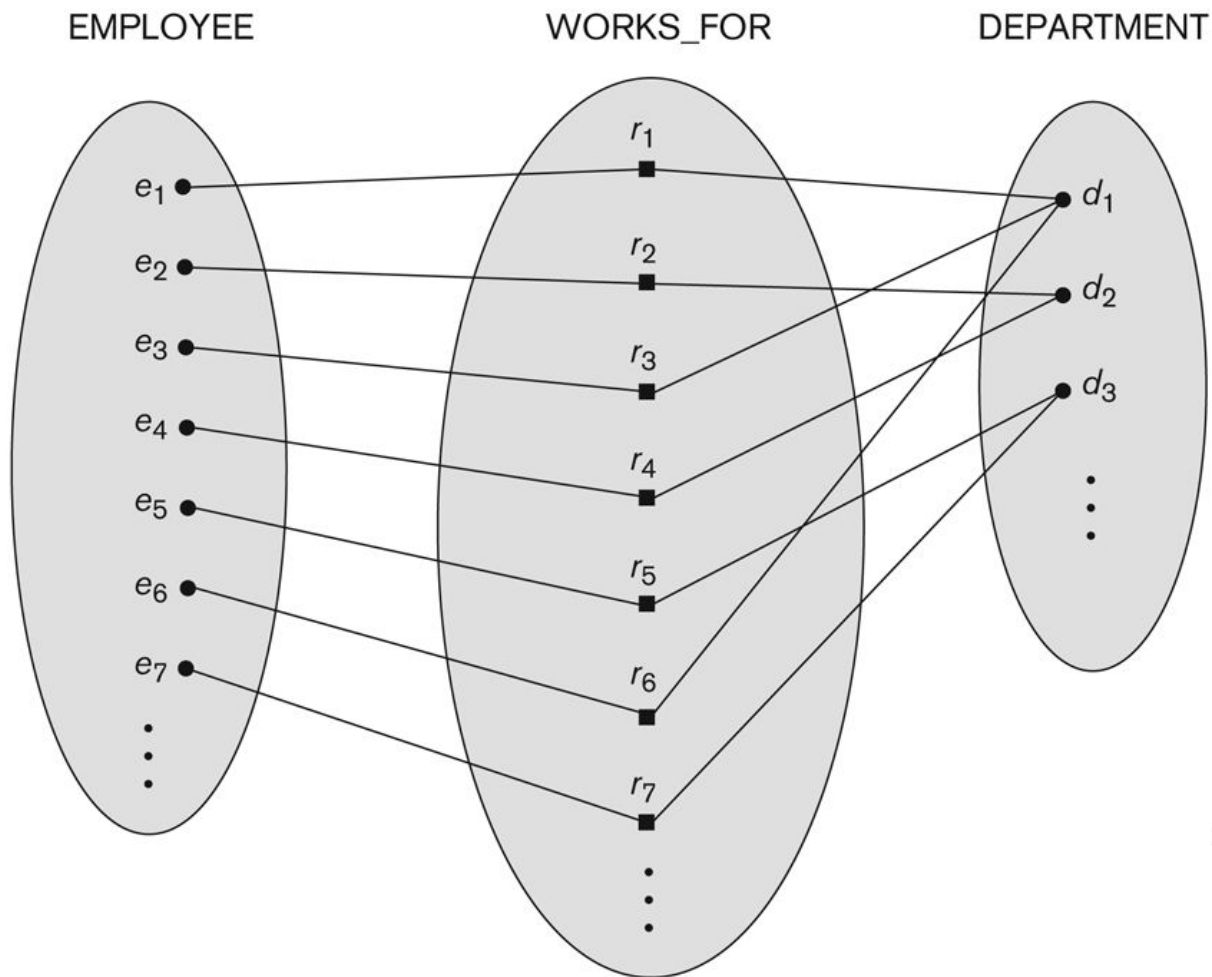


Figure 3.9

Some instances in the WORKS_FOR relationship set, which represents a relationship type WORKS_FOR between EMPLOYEE and DEPARTMENT.

MANY-TO-MANY (M:N) RELATIONSHIP

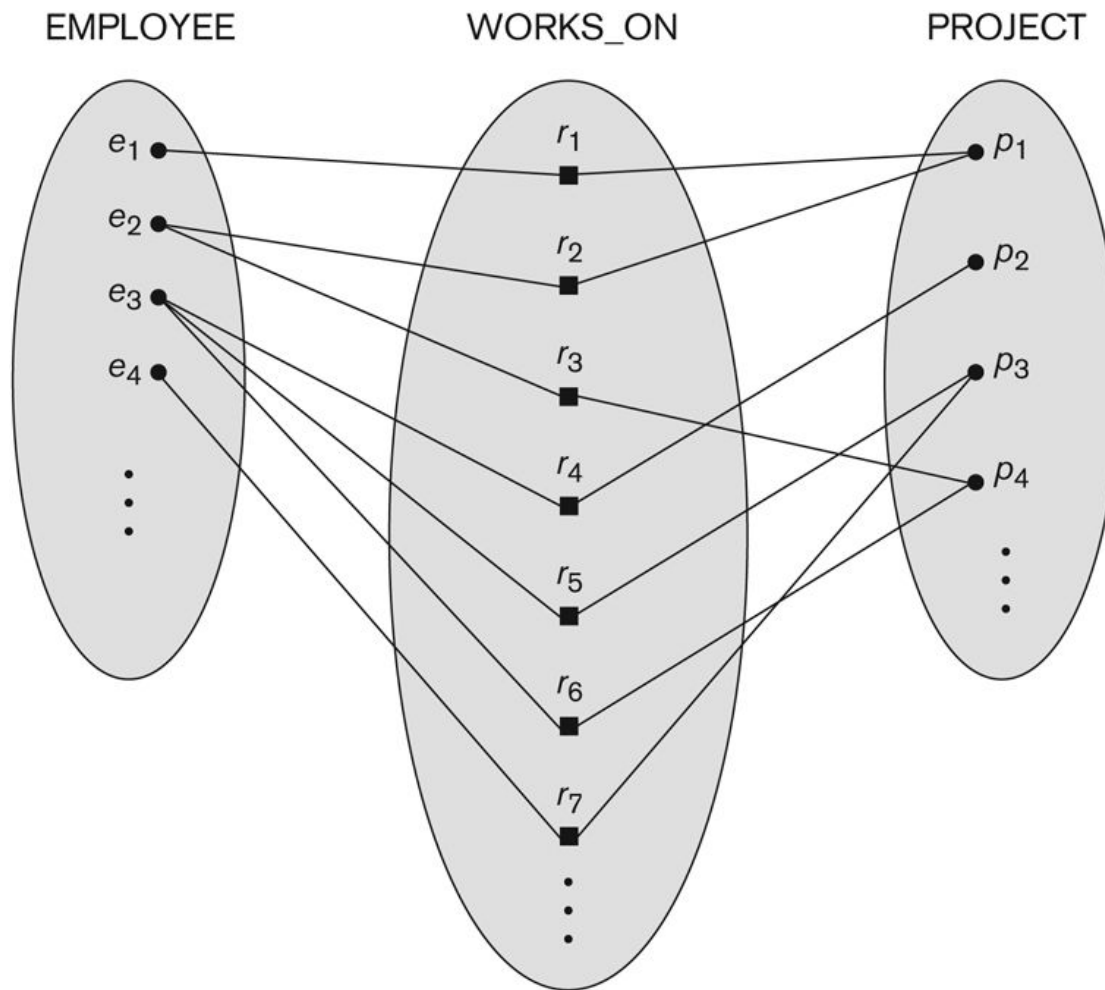


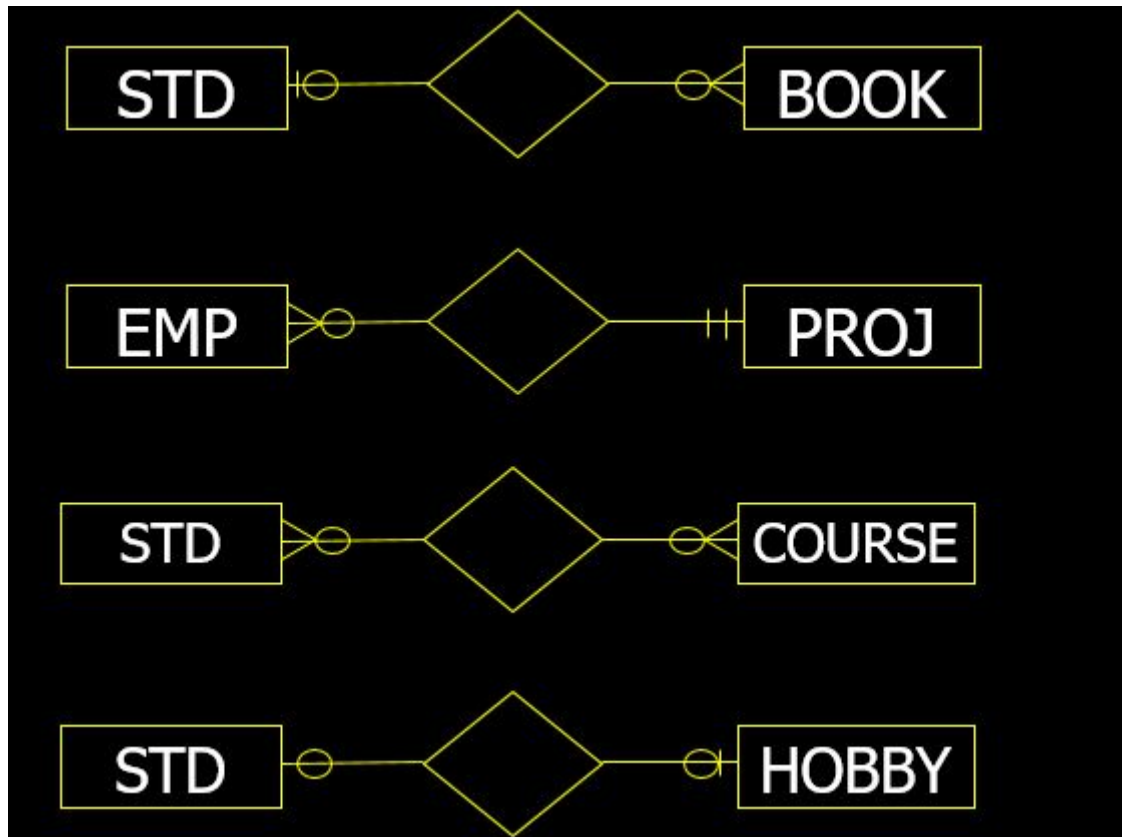
Figure 3.13
An M:N relationship,
WORKS_ON.

MINIMUM CARDINALITY

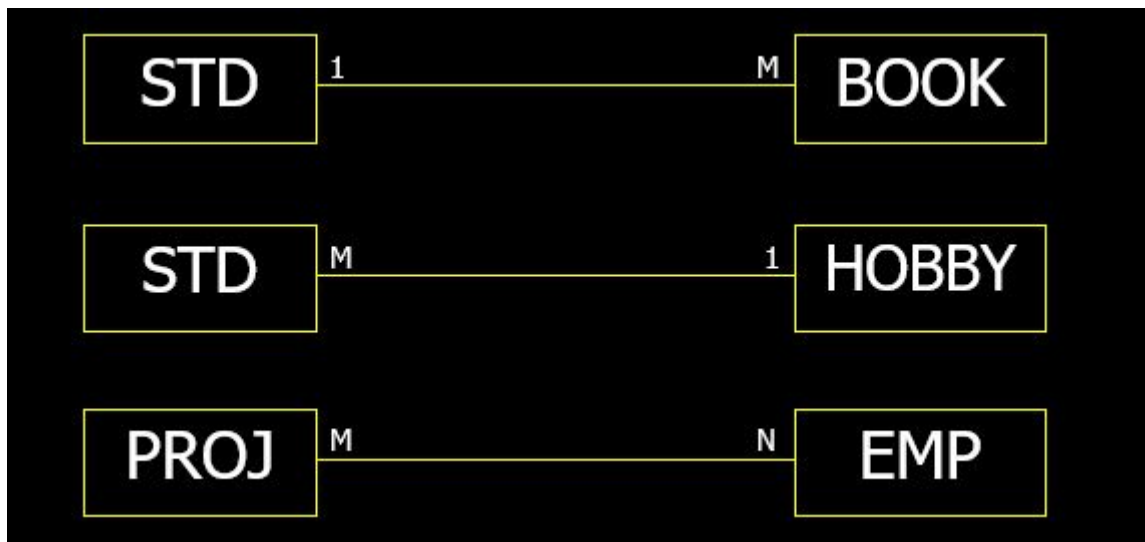
- Determines whether the link is compulsory or optional
- Important, since it effects the implementation



CARDINALITY EXAMPLE(Crow's Foot NOTATION)

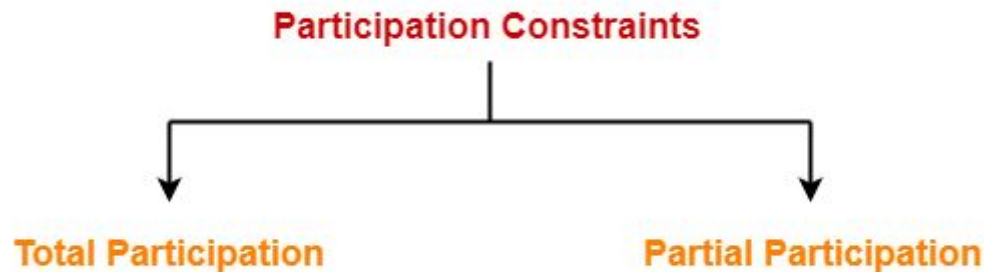


OTHER NOTATIONS(CHEN)



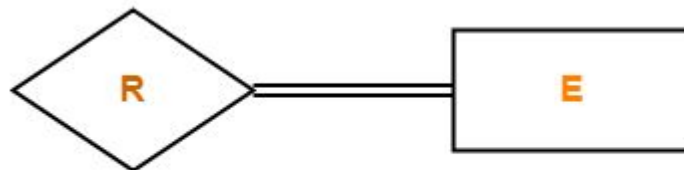
PARTICIPATION CONSTRAINTS-

- Participation constraints define the least number of relationship instances in which an entity must compulsorily participate.
- There are two types of participation constraints-



TOTAL PARTICIPATION-

- It specifies that each entity in the entity set must compulsorily participate in at least one relationship instance in that relationship set.
- That is why, it is also called as **mandatory participation**.
- Total participation is represented using a double line between the entity set and relationship set.



Total Participation



EXAMPLE

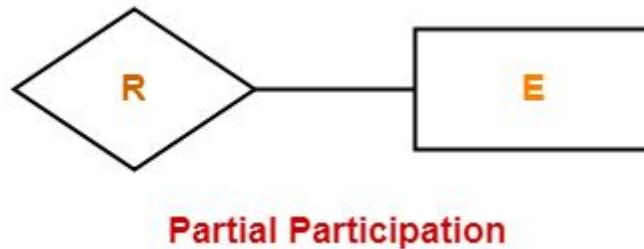


- Double line between the entity set “Student” and relationship set “Enrolled in” signifies total participation.
- It specifies that each student must be enrolled in at least one course.



PARTIAL PARTICIPATION-

- It specifies that each entity in the entity set may or may not participate in the relationship instance in that relationship set.
- That is why, it is also called as **optional participation**.
- Partial participation is represented using a single line between the entity set and relationship set.



EXAMPLE



- Single line between the entity set “Course” and relationship set “Enrolled in” signifies partial participation.
- It specifies that there might exist some courses for which no enrollments are made.



RELATIONSHIP BETWEEN CARDINALITY AND PARTICIPATION CONSTRAINTS-

- Minimum cardinality tells whether the participation is partial or total.
- If minimum cardinality = 0, then it signifies partial participation.
- If minimum cardinality = 1, then it signifies total participation.
- Maximum cardinality tells the maximum number of entities that participates in a relationship set.



ER DIAGRAM – RELATIONSHIP TYPES ARE: WORKS_FOR, MANAGES, WORKS_ON, CONTROLS, SUPERVISION, DEPENDENTS_OF

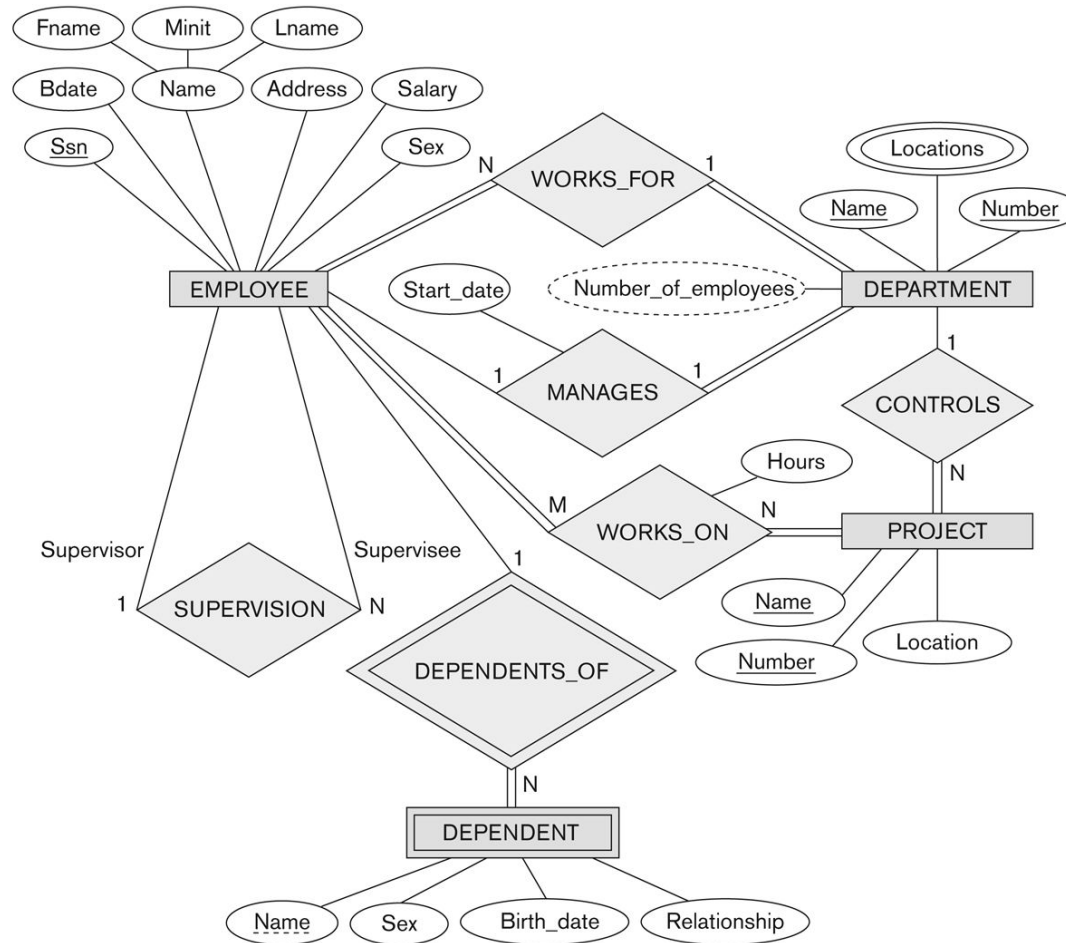


Figure 3.2

An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

STEPS TO DRAW AN ER DIAGRAM

- Get problem description
- Define Entities
- Add Attributes
- Specify Key, multiple, composite attributes
- Add Relations
- Specify Cardinality, total/partial relations
- Iterate



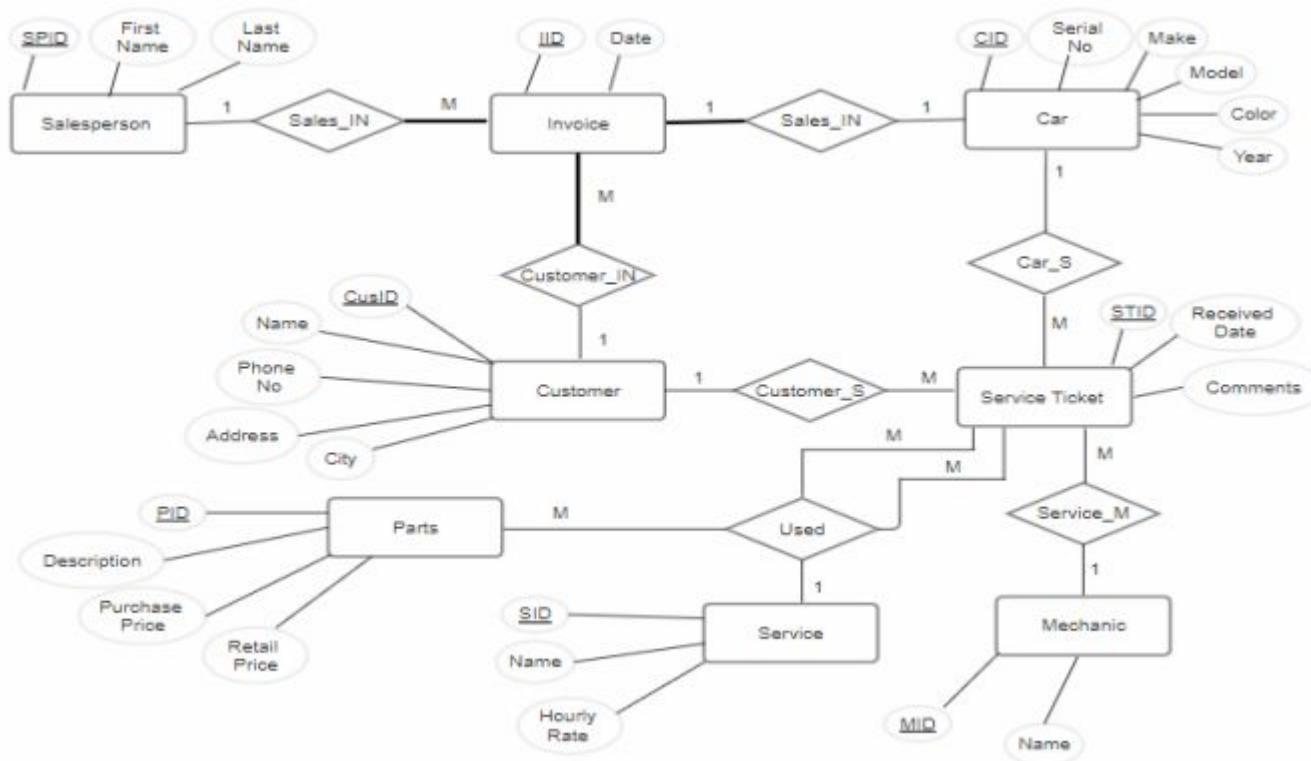
STEPS TO DRAW AN ER DIAGRAM

- Essential to further design, but often given little care:
- Is an entity a weak entity (key?)
- Multivalued attributes
- Derived attribute
- Total/partial participation
- Cardinality ratio



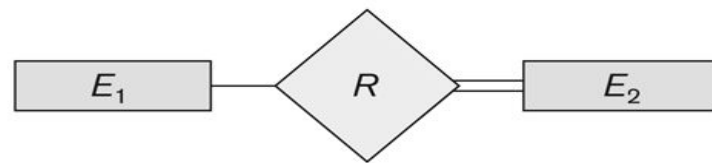
PROBLEM

The dealership sells both new and used cars, and it operates a service facility. A salesperson (SPID, Last Name, First Name) may sell many cars, but each car (CID, Serial No, Make, Model, Color, Year) is sold by only one salesperson. A customer (CusID, Name, Phone No, Address, City) may buy many cars, but each car is bought by only one customer. A salesperson writes a single invoice (IID, Date) for each car he or she sells. A customer gets an invoice for each car he or she buys. A customer may come in just to have his or her car serviced; that is, a customer need not buy a car to be classified as a customer. When a customer takes one or more cars in for repair or service (SID, Name, Hourly Rate), one service ticket (STID, Received Date, Comments) is written for each car. The car dealership maintains a service history for each of the cars serviced. The service records are referenced by the car's serial number. A car brought in for service can be worked on by many mechanics (MID, Name) and each mechanic may work on many cars. A car that is serviced may or may not need parts (PID, Description, Purchase price, Retail price) adjusting a carburetor or cleaning a fuel injector nozzle does not require providing new parts.

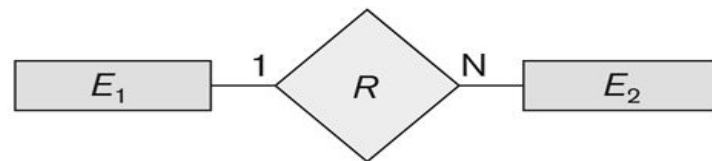


ALTERNATIVE (MIN, MAX) NOTATION FOR RELATIONSHIP STRUCTURAL CONSTRAINTS:

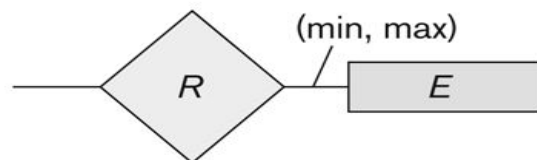
- Specifies that each entity e in E participates in at least min and at most max relationship instances in R
- Default(no constraint): $min=0$, $max=n$
- Must have $min \leq max$, $min \geq 0$, $max \geq 1$
- Derived from the knowledge of mini-world constraints



Total Participation of E_2 in R

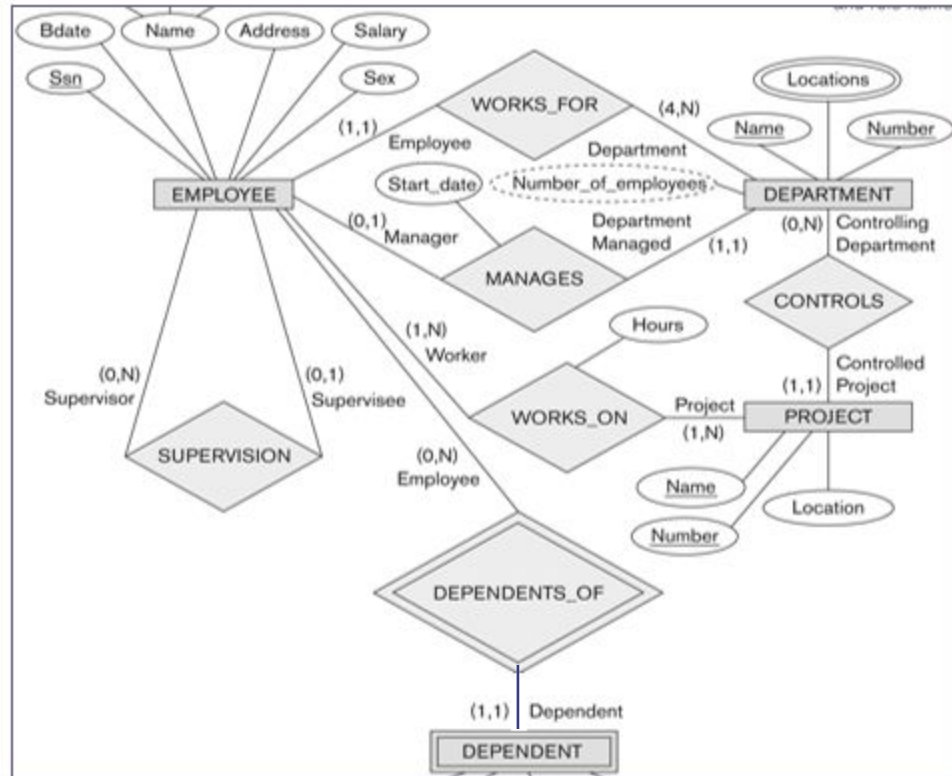


Cardinality Ratio 1: N for $E_1:E_2$ in R



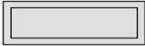
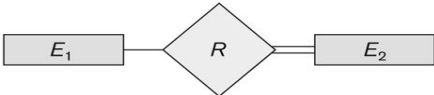
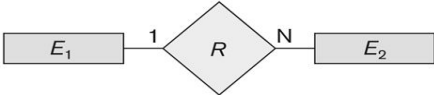
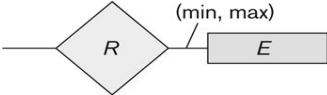
Structural Constraint (min, max)
on Participation of E in R

MIN, MAX - NOTATION



NOTATION FOR ER DIAGRAMS

Figure 3.14
Summary of the
notation for ER
diagrams.

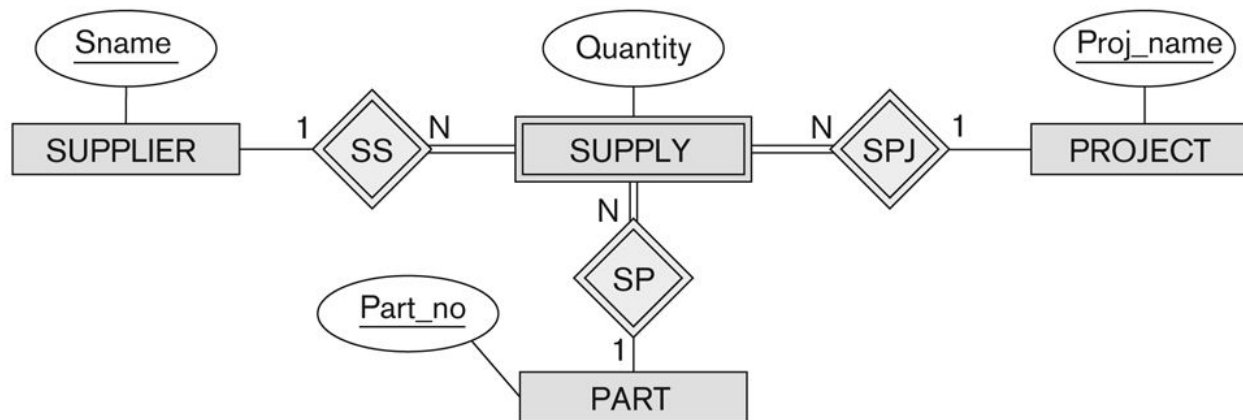
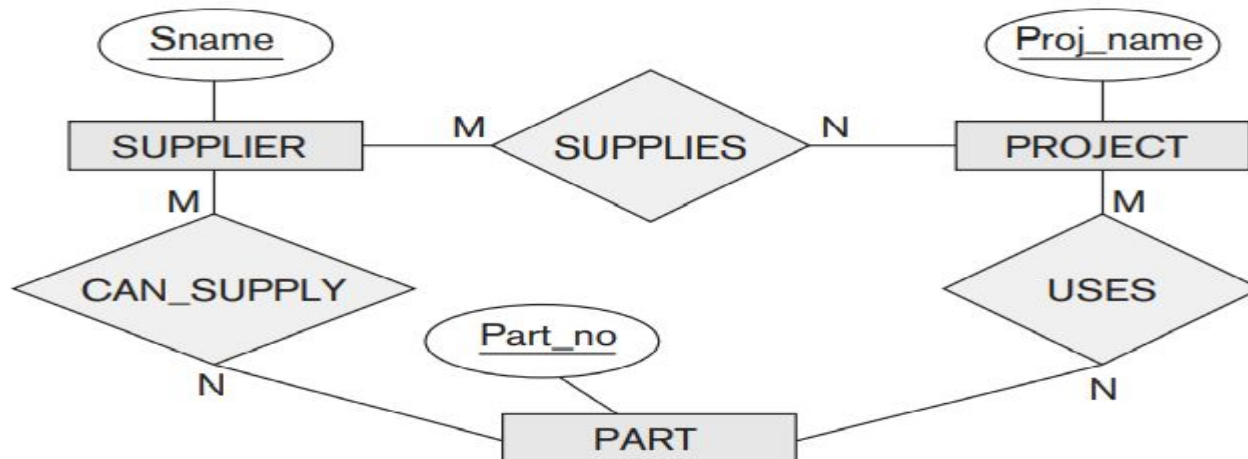
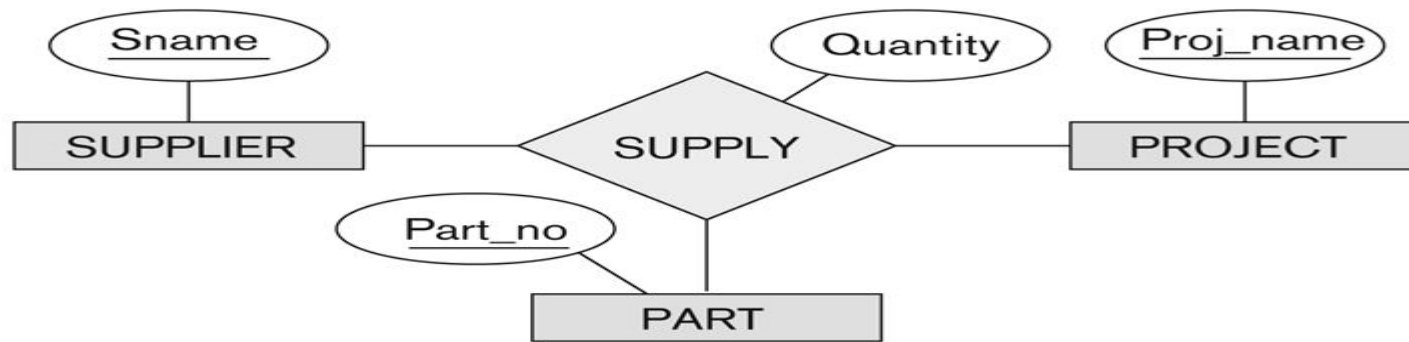
Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1: N for $E_1:E_2$ in R
	Structural Constraint (min, max) on Participation of E in R

RELATIONSHIPS OF HIGHER DEGREE

- Relationship types of degree 2 are called binary
- Relationship types of degree 3 are called ternary and of degree n are called n -ary
- Constraints are harder to specify for higher-degree relationships ($n > 2$) than for binary relationships

N-ARY RELATIONSHIPS ($N > 2$)

- Three binary relationships represents different information than a single ternary relationship
- In some cases, a ternary relationship can be represented as a weak



ALTERNATIVE DIAGRAMMATIC NOTATION

- ER diagrams is one popular example for displaying database schemas
- UML class diagrams is representative of another way of displaying ER concepts

OTHER ALTERNATIVE DIAGRAMMATIC NOTATIONS

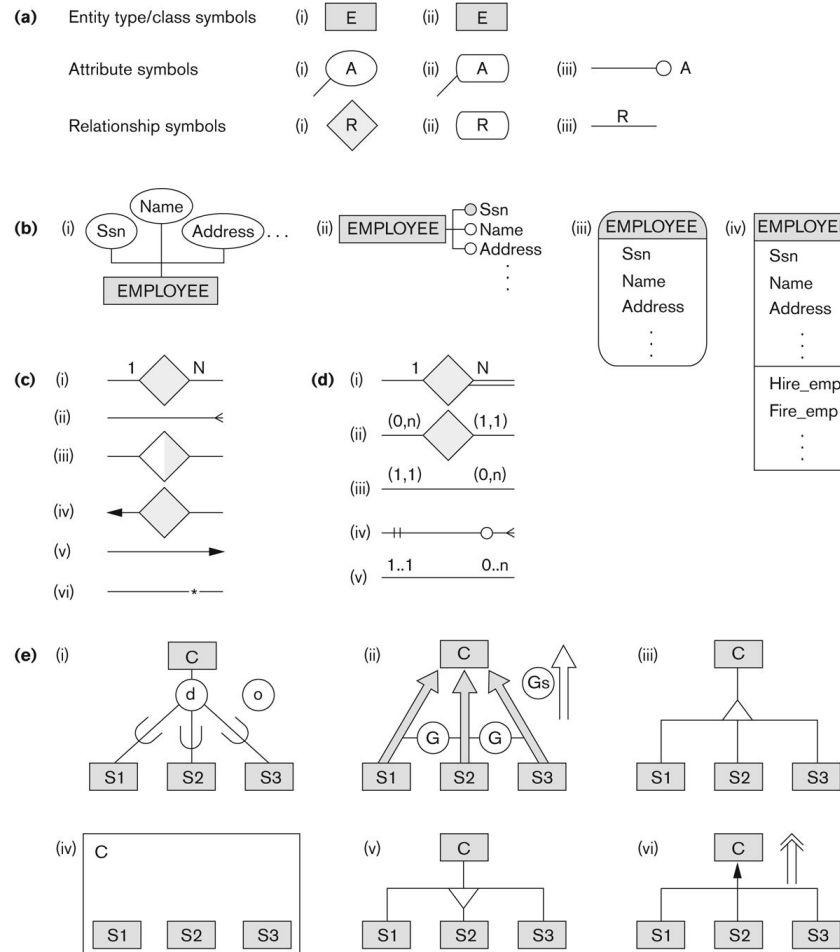


Figure A.1

Alternative notations. (a) Symbols for entity type/class, attribute, and relationship. (b) Displaying attributes. (c) Displaying cardinality ratios. (d) Various (min, max) notations. (e) Notations for displaying specialization/generalization.